PROGRAM 1

TRAFFIC LIGHT CONTROLLER

SOURCE CODE:

```
/ /*
Formal Design of Embedded Controller Using FSM Traffic Light Controller

Arduino UNO + Proteus
*/

enum TrafficState {
   STATE_NS_GREEN,
   STATE_NS_YELLOW,
   STATE_EW_GREEN,
   STATE_EW_YELLOW
};

TrafficState currentState = STATE_NS_GREEN;

// North-South traffic signal pins
const int NS_RED_PIN = 2;
const int NS_YELLOW_PIN = 3;
const int NS_GREEN_PIN = 4;

// East-West traffic signal pins
const int EW_RED_PIN = 5;
const int EW_YELLOW_PIN = 6;
const int EW_GREEN_PIN = 7;

void allOff() {
   digitalWrite(NS_RED_PIN, LOW);
```

```
    digitalWrite(NS_YELLOW_PIN, LOW);
    digitalWrite(NS_GREEN_PIN, LOW);

    digitalWrite(EW_RED_PIN, LOW);
    digitalWrite(EW_YELLOW_PIN, LOW);
    digitalWrite(EW_GREEN_PIN, LOW);
}

void enterState(TrafficState state) {
    allOff();

    switch (state) {

        case STATE_NS_GREEN:
            digitalWrite(NS_GREEN_PIN, HIGH);
            digitalWrite(EW_RED_PIN, HIGH);
            Serial.println("S0: NS GREEN - EW RED");
            break;

        case STATE_NS_YELLOW:
            digitalWrite(NS_YELLOW_PIN, HIGH);
            digitalWrite(EW_RED_PIN, HIGH);
            Serial.println("S1: NS YELLOW - EW RED");
            break;

        case STATE_EW_GREEN:
            digitalWrite(NS_RED_PIN, HIGH);
            digitalWrite(EW_GREEN_PIN, HIGH);
            Serial.println("S2: NS RED - EW GREEN");
            break;
```

```
        case STATE_EW_YELLOW:
            digitalWrite(NS_RED_PIN, HIGH);
            digitalWrite(EW_YELLOW_PIN, HIGH);
            Serial.println("S3: NS RED - EW YELLOW");
            break;
    }
}

void setup() {
    pinMode(NS_RED_PIN, OUTPUT);
    pinMode(NS_YELLOW_PIN, OUTPUT);
    pinMode(NS_GREEN_PIN, OUTPUT);

    pinMode(EW_RED_PIN, OUTPUT);
    pinMode(EW_YELLOW_PIN, OUTPUT);
    pinMode(EW_GREEN_PIN, OUTPUT);

    Serial.begin(9600);

    enterState(currentState);
}

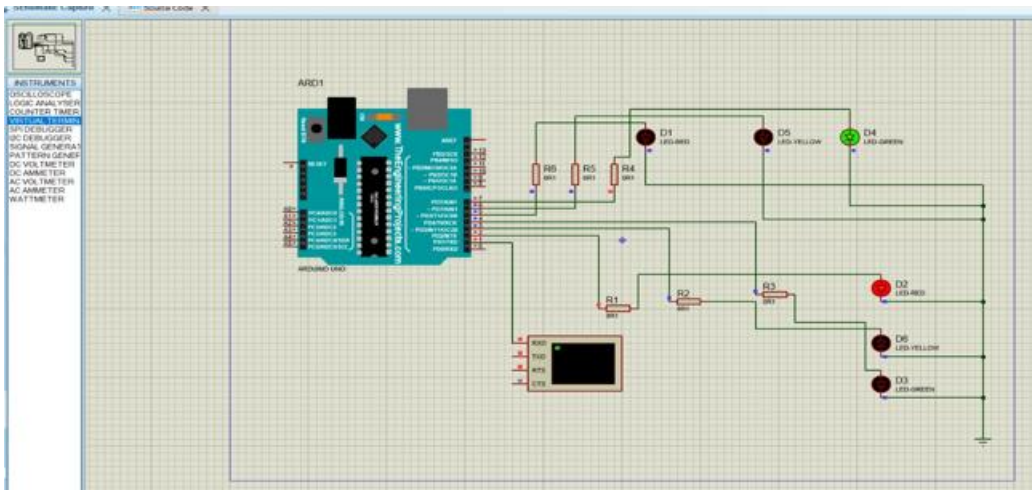
void loop() {
    switch (currentState) {

        case STATE_NS_GREEN:
            delay(10000); // 10 seconds
            currentState = STATE_NS_YELLOW;
            break;
```

```
        case STATE_NS_YELLOW:
            enterState(STATE_NS_YELLOW);
            delay(3000); // 3 seconds
            currentState = STATE_EW_GREEN;
            break;

        case STATE_EW_GREEN:
            enterState(STATE_EW_GREEN);
            delay(10000); // 10 seconds
            currentState = STATE_EW_YELLOW;
            break;

        case STATE_EW_YELLOW:
            enterState(STATE_EW_YELLOW);
            delay(3000); // 3 seconds
            currentState = STATE_NS_GREEN;
            enterState(currentState);
            break;
    }
}
```



OBSERVATION:

- The traffic light controller is implemented using an Arduino UNO microcontroller and a Finite State Machine (FSM) approach.
- Four operating states are defined: North-South Green, North-South Yellow, East-West Green, and East-West Yellow.
- The North-South green signal remains ON for 10 seconds, allowing vehicles to move in that direction.
- The North-South yellow signal remains ON for 3 seconds to indicate that the signal is about to change.
- The East-West green signal remains ON for 10 seconds, allowing vehicles to move in the perpendicular direction.
- The East-West yellow signal remains ON for 3 seconds before the controller returns to the North-South green state.
- The `digitalWrite()` function controls the traffic signal LEDs by switching them ON and OFF according to the current state.
- The `switch-case` statement manages the FSM states and determines the sequence of traffic light operations.
- The Serial Monitor displays the current traffic light state, helping users observe and verify the state transitions.
- The system is simulated using Proteus and demonstrates FSM-based control, automatic state transitions, and sequential traffic signal operation using Arduino programming.